

Hyunseung Hwang

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Education

PhD	KAIST , Electrical Engineering • Advisor: Steven Euijong Whang (Data Intelligence Lab) • Dissertation: Enhancing Explainability in Machine Learning — Explainable Clustering and Explanation Multiplicity	Daejeon, South Korea Sept 2021 – Aug 2026
MS	KAIST , Electrical Engineering	Daejeon, South Korea Sept 2019 – Aug 2021
BS	KAIST , Computer Science	Daejeon, South Korea Sept 2013 – Aug 2019

Experience

New York University (NYU), Center for Responsible AI , Postdoctoral Researcher • Conducting advanced research on the interpretability and safety of machine learning models under the supervision of Prof. Julia Stoyanovich. • Stress-testing explanation models to ensure safe, confident, and defensible outputs for end-users and stakeholders in high-stakes domains.	Brooklyn, NY, USA June 2026 – present 1 month
Global Frontier Lab , Research Collaborator • Collaborated with Julia Stoyanovich's Responsible AI group at NYU. • Investigated the sensitivity and robustness of AI explainability techniques to deliver safe, confident explanations to stakeholders.	Daejeon, South Korea Sept 2025 – Dec 2025 4 months
Naver , Research Intern • Implemented community detection (Girvan–Newman) on PySpark to cluster unsupervised data for video recommendation.	Seongnam, South Korea June 2019 – Aug 2019 3 months
Kiswe Mobile , Software Engineering Intern • Built Android features for a live video-streaming service using Android Studio. • Updated the application to meet Android Oreo (8.0+) requirements.	Murray Hill, NJ, USA June 2018 – Aug 2018 3 months

Publications

SHAP-based Explanations are Sensitive to Feature Representation Shows that routine data-engineering choices can manipulate SHAP feature importance for the same model, with direct implications for model risk and fair lending. Hyunseung Hwang, Andrew Bell, Joao Fonseca, Venetia Pliatsika, Julia Stoyanovich, Steven Euijong Whang 10.1145/3715275.3732105 (ACM Conference on Fairness, Accountability, and Transparency (FAccT))	June 2025
XClusters: Explainability-first Clustering Makes explainability a first-class objective of clustering via a difference-of-monotonic-functions formulation; evaluated on a real credit-card dataset. Hyunseung Hwang, Steven Euijong Whang arxiv.org/abs/2209.10956 (AAAI Conference on Artificial Intelligence (AAAI))	June 2023

Open-World COVID-19 Data Visualization

Mar 2021

On the difficulty of producing useful visualizations in an open-world setting where the relevant datasets are not known in advance.

Hyunseung Hwang, Steven Euijong Whang

[10.1007/978-3-030-71055-2_8](https://doi.org/10.1007/978-3-030-71055-2_8) (VLDB Workshop on Data Management and Analytics for Medicine and Healthcare (DMAH))

Skills

Research: Explainable AI (XAI), SHAP / feature attribution, Explanation multiplicity and robustness, Fairness and responsible AI, Explainable clustering

Technical: Python, PySpark, Machine learning, Data visualization

Languages

Korean: Native speaker

English: Fluent